

Jia-Fong Yeh (葉佳峯)

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PhD in Computer Science at NTU with extensive research and practical experience in Machine Learning and Robotics. Dedicated to addressing core challenges in real-world robotic applications, with publications at top venues such as AAAI, NeurIPS, and ICLR. Demonstrated leadership as a team lead of 10+ researchers in the university lab and through international collaboration with Sony. Recognized for research excellence and practical contributions with multiple national and academic awards.

Working Experiences

Senior Associate Researcher, Delta Robotics Innovation Center

Taipei, Taiwan

- Develop digital twin and scene reconstruction pipelines *Aug. 2025 – now*
- Be the PI or involved in robotics projects collaborated with university labs

ML Research Intern, Sony Group Corporation

Tokyo, Japan

- Mentor: Dr. Wang Zhao and Mr. Shingo Takamatsu *Oct. 2023 – Mar. 2024*
- Develop projects related to *Reinforcement Learning with LLM/VLM guidance*

Education

National Taiwan University (NTU)

Taipei, Taiwan

PH.D. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

Sept. 2019 – July 2025

- GPA: 4.22/4.3
- Advisor: Prof. Winston H. Hsu

National Taiwan Normal University (NTNU)

Taipei, Taiwan

B.S. & M.S. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING

Sept. 2013 – June 2019

- GPA: 3.74/4.3 & 4.3/4.3
- Advisor: Prof. Tsung-Che Chiang and Prof. Shun-Shii Lin

Selected Research Projects

My research focuses on advancing **policy learning**, addressing **three major challenges**: (1) the complexity of the environment, (2) the lack of reward signals, and (3) the safety concerns on policy's behavior, as described below:

Learning Multi-stage Manipulation Tasks from Few Demonstrations | AAAI 2022

- Introduced an **attention-based** policy that learns from demonstrations on complex tasks
- Outperformed all baselines in **6 out of 8 cases**, with interpretable visualization results

Vision-Instruction Correlation Reward Generation | ICLR 2025

- Developed a hierarchical reward model using both **LLM** and **VLM** techniques
- Demonstrated a **43%** improvement over the strongest baseline in **long-horizon tasks**

Monitoring Policy's Behaviors in Unseen Environments | NeurIPS 2024

- Proposed novel **contrastive losses** to enhance monitoring for detecting erroneous actions
- Achieved the best results in **17 out of 21 cases**, with extensive ablations and visualizations

Skills

- **Programming Languages:** C/C++, Python, C#
- **ML Infrastructure/Packages:** PyTorch, TensorFlow, Weights&Biases, scikit-learn, AWS EC2
- **Robot Simulation:** Pybullet, Isaac Sim, Coppeliassim, ROS 2
- **Others** – Git, Docker, HTML, CSS, mysql, OpenMP
- **Selected Projects** –WebGL renderer [\[demo\]](#), Dark Chess AI, JPEG Decoder, Snake Game AI

Academic Services

Conference Committee

- **Area Chair:** ICLR'26
- **Reviewer:** CVPR (*outstanding reviewer* in CVPR'25), NeurIPS, ICML, ICLR, ICCV, AAAI, WACV, ICASSP, ICDL, ICMLW, CVPRW

Selected Awards

- **Dissertation/Thesis Awards:** 2025 IPPR dissertation competition [\[link\]](#), 2025 TSC dissertation competition [\[link\]](#), 2019 ORSTW thesis competition [\[link\]](#)
- **Research Awards:** 2025 Hon Hai Technology Award [\[link\]](#), 2025 NTUEE-1975 Technology Research Innovation Award [\[link\]](#), 2024 NeurIPS Scholar Award
- **PhD Scholarship/Membership:** 2025 NTU Presidential Award for Graduate Students [\[link\]](#), 2025 Phi Tau Phi Honorary Member [\[link\]](#), 2022 NOVATEK PhD Scholarship [\[link\]](#)
- **Other Competitions:** 2018 ITSA Annual Collegiate Programming Contest [\[link\]](#), 2016-2022 Computer Chess Competition (16 medals)

Selected First-author Works | citations: 460+ | h-index: 8 | i-10 index: 6

- VICtoR: Learning Hierarchical Vision-Instruction Correlation Rewards for Long-horizon Manipulation
ICLR 2025 | co-first work | AR: 32.08% | [\[PDF\]](#)
- AED: Adaptable Error Detection for Few-shot Imitation Policy
NeurIPS 2024 | AR: 25.8% | [\[PDF\]](#)
- Stage Conscious Attention Network (SCAN): A Demonstration-conditioned Policy for Few-shot Imitation
AAAI 2022 | co-first work | AR: 15% | [\[PDF\]](#)
- Large Margin Mechanism and Pseudo Query Set on Cross-Domain Few-Shot Learning
CVPRW Report 2020 | citation: 20+ | [\[PDF\]](#)
- Modified L-SHADE for Single Objective Real-Parameter Optimization

CEC 2019 | citation: 20+ | [\[PDF\]](#)

- Snake Game AI: Movement Rating Functions and Evolutionary Algorithm-based Optimization

TAAI 2016 | [\[PDF\]](#)

I also have other publications in *IEEE TMM*, *IEEE TAI*, *AAAI*, *ICCV*, *EMNLP*, *CoRL*, *ICRA*, *ICASSP*, *BMVC*, *ICIP*, *ECCVW*, and *NeurIPSW*, please find them from my [Google Scholar](#).